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Faculty of Computer Studies
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***< Development and Implementation of an Advanced Firewall
Using NGFW with Cypher Security Technologies >***

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TM471: Final Year Project, 2024/2025

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1. Abstract

This Project creates a working implementation of an OPNsense-based NGFW (Next Generation Firewall) that addresses today's security issues, which cannot be handled by traditional firewalls (i.e. checking for incoming/outgoing traffic) with a number of the following features, IDS/IPS, IPsec VPN, DNS Filtering. These features improve detection of attacks, provide extra layers of security for communications, and improve control over who has access to your network. After testing with various scans and measuring the effectiveness of blocked domains, we have demonstrated that the NGFW provides better visibility and greater protection than traditional methods and allows for far better control over the flow of data over your network.

2. Acknowledgement

I would like to extend my heartfelt thanks to everyone who offered their support and assistance during this journey. I am especially grateful to my supervisor, Dr. Hussein Albazzar, for his unwavering guidance, insightful advice, and constant encouragement at every stage of this project. Finally, I owe my deepest gratitude to my family for their understanding, boundless moral support, and unshakable faith in me.

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Chapter 4: Design, Implementation, and Testing

Introduction

This chapter outlines the design, implementation, and testing of the NGFW setup using OPNsense. The virtual lab included HQ and Branch firewalls, an ISP router, and a Kali machine for security testing. Key features such as IPsec, IDS/IPS, firewall rules, and DNS filtering were configured and validated. The results show how the NGFW handled scanning attempts, secured communication, and supported normal network operations.

System Architecture and Integration Overview

The system uses a virtual network with an HQ and a Branch connected through an ISP router and a site-to-site IPsec tunnel. Each site runs an OPNsense firewall for routing, filtering, and security functions. A Kali machine was added for controlled attack testing, along with virtual PCs to simulate users. The setup demonstrates how an NGFW combines firewalling, VPN, IDS/IPS, and DNS filtering to protect and monitor network traffic.

3.1 Hardware Setup and software setup

3.1.1 Firewall Design (HQ & Branch)

The HQ and Branch firewalls were configured with different security levels to compare protected and unprotected environments. The HQ used full NGFW features, including IDS/IPS and DNS filtering, while the Branch relied only on basic firewall rules. This setup made it easy to observe how the HQ detects and blocks attacks like *nmap -A*, while the Branch does not, clearly showing the value of enabling advanced security controls.

3.1.1.1 DHCP Configuration.

Both the HQ and Branch firewalls were configured with DHCP services to automatically assign IP addresses to devices within their respective LAN networks. Each firewall used its own DHCP range that matched its local subnet, ensuring that clients could join the network without manual configuration. This setup simplified device management and allowed end-users, test machines, and VPCS clients to obtain IP settings such as gateway and DNS automatically from the firewall. The DHCP service played an essential role in enabling communication inside each site and supporting the overall NGFW topology.

HQ-DHCP range : 192.168.10.50 – 192.168.10.100.

BRANCH-DHCP range : 192.168.20.50 – 192.168.20.100

3.1.1.2 Network Access Control

Network access control was enforced through firewall rules on both OPNsense systems. The HQ used stricter rules and IDS/IPS inspection, while the Branch kept minimal rules to allow comparison during testing. This ensured controlled traffic flow and highlighted the difference between protected and unprotected environments.

3.1.1.3 WAN/LAN Configuration

Both the HQ and Branch firewalls were configured with separate WAN and LAN interfaces to establish clear network boundaries and support routing between internal and external segments. The WAN interfaces of both firewalls connect to the simulated ISP router, which enables communication between the two sites and allows the IPsec VPN tunnel to be established. Each WAN interface was assigned an IP address appropriate to its network, ensuring proper reachability through static routes and gateway settings.

The LAN interfaces provide connectivity to local devices within the HQ and Branch networks. Each LAN network uses its own subnet, allowing the firewalls to manage internal traffic and apply local security rules. The HQ LAN is also monitored by IDS/IPS to inspect internal traffic, while the Branch LAN operates without inspection for comparison purposes.

This separation of WAN and LAN provides a clear structure for traffic flow, supports secure site-to-site communication, and allows testing of security features under different conditions.

Table 1:HQ Firewall Interfaces

Interface	IP Address	Subnet	Description
WAN	192.168.163.10	/24	Connected to ISP router
LAN	192.168.10.1	/24	HQ Internal network
VPN (IPsec)	---	Tunnel	Site-to-site connectivity

Table 2:Branch Firewall Interfaces

Interface	IP Address	Subnet	Description
WAN	192.168.71.10	/24	Connected to ISP router
LAN	192.168.20.1	/24	Branch Internal network
VPN (IPsec)	---	Tunnel	Site-to-site connectivity

3.1.1.4 VPN Configuration (IPsec Phase 1 & Phase 2)

The site-to-site VPN between the HQ and Branch was implemented using IPsec to provide a secure communication channel across the simulated external network. Both firewalls were configured with matching Phase 1 and Phase 2 settings to ensure the tunnel could be established successfully.

In Phase 1, the connection method was set to Start on traffic, using IKEv2 as the key exchange version and IPv4 as the protocol. The WAN interface was selected on both sides because the VPN traffic passes through the ISP router. Authentication was configured with Mutual PSK, and the same pre-shared key (Hamza-VPN-20464317) was entered on both devices. For encryption and integrity, AES-256 and SHA-256 were used along with DH Group 14 (2048-bit), providing a strong cryptographic profile.

Edit pre-shared-key

Local Identifier	192.168.163.10
Remote Identifier	192.168.71.10
Pre-Shared Key	Hamza-VPN-20464317
Type	PSK
Description	

Figure 1: pre-shared-key

Connections
HQ-to-Branch
Pools

General settings

advanced mode	☐
enabled	☒
Proposals	aes256-sha256-modp2048 [DH14] Clear All Select All
Version	IKEv2
MOBIKE	☐
Local addresses	192.168.163.10 Clear All Copy Text
Remote addresses	192.168.71.10 Clear All Copy Text
DPD delay (s)	30
Pools	Nothing selected Clear All Select All
Description	HQ-to-Branch

Figure 2: HQ-IPsec-Phase1

Connection method	Start on traffic
Key Exchange version	V2
Internet Protocol	IPv4
Interface	WAN
Remote gateway	192.168.163.10
Dynamic gateway	☐ Allow any remote gateway to connect
Description	Branch-to-HQ
Phase 1 proposal (Authentication)	
Authentication method	Mutual PSK
My identifier	My IP address
Peer identifier	Peer IP address
Pre-Shared Key	Hamza-VPN-20464317

Figure 3: Branch-IPsec-Phase1



Encryption algorithm	AES
	256
Hash algorithm	SHA256
DH key group	14 (2048 bits)
Lifetime	28800

Figure 4: Branch-IPsec-Phase1

Phase 2 defined the actual networks that would communicate through the tunnel. On the HQ firewall, the local network was set to the HQ LAN (192.168.10.0/24), and the remote network was set to the Branch LAN (192.168.20.0/24). The Branch firewall used the opposite configuration. ESP was selected as the Phase 2 protocol with AES-256 and SHA-256 for encryption and hashing, along with DH Group 14 for PFS.

Edit Child

advanced mode

enabled

☒

Connection

HQ-to-Branch

Mode

Tunnel

Policies

☒

Start action

Start

DPD action

Clear

Reqid

ESP proposals

aes256-sha256-modp2048 [DH14]

Clear All

Select All

Local

192.168.10.0/24

Clear All

Copy

Text

Remote

192.168.20.0/24

Clear All

Copy

Text

Description

HQ-to-BranchP2

Figure 5: HQ-IPsec-Phase2

Mode	Tunnel IPv4
Description	Branch-to-HQ-P2
Local Network	
Type	LAN subnet
Address:	
	/
	32
Remote Network	
Type:	Network
Address:	192.168.10.0
	/
	24
Phase 2 proposal (SA/Key Exchange)	
Protocol	ESP
Encryption algorithms	☒ AES
	256 bits

Figure 6: Branch-IPsec-Phase2

After applying the configurations on both sides, the IPsec tunnel came up successfully, allowing devices in the HQ to communicate securely with devices in the Branch.

The screenshot displays the OPNsense IPsec tunnel configuration interface. It is divided into two main sections: Phase 1 and Phase 2.

Phase 1: This section shows the tunnel's status and basic configuration. The table below represents the data shown in the interface.

Status	Connection	Version	Local ID	Local IP	Remote ID	Remote IP	Time	Bytes in	Bytes out
☒	HQ-to-Branch	IKEv2	192.168.163.10	192.168.163.10	192.168.71.10	192.168.71.10	2		

Phase 2: This section shows the tunnel's subnets and state. The table below represents the data shown in the interface.

Local subnets	spi-in	spi-out	Remote subnets	State	Time	Bytes in	Bytes out
☐ 192.168.10.0/24	cb86282d	cd118a23	192.168.20.0/24	INSTALLED	2		

Figure 7: IPsec tunnel

3.1.1.5 Routing & Static Routes

Routing between the HQ and Branch networks required adding static routes on the intermediate devices to ensure traffic could reach the correct destination networks. Since both firewalls are connected through an ISP router, each device needed a route that points to the other site's LAN through the proper next-hop address.

On the ISP router, static routes were added for the HQ LAN (192.168.10.0/24) and the Branch LAN (192.168.20.0/24) so that traffic could be forwarded to the correct firewall. Similarly, the Kali router used additional routes to allow attack traffic to reach the HQ firewall and to simulate external scanning behavior.

The firewalls themselves relied primarily on their default gateway and the IPsec tunnel for routing. Once the tunnel was established, the HQ and Branch were able to automatically exchange packets using the encrypted path without requiring extra static routes on the OPNsense devices. These routing steps ensured stable communication and allowed IPS/IDS testing traffic to reach the HQ firewall as intended.

Edit route

☒ Disabled

Network Address

Gateway

Description

Figure 8: HQ-route to kali network

Edit route

☐ Disabled

Network Address

Gateway

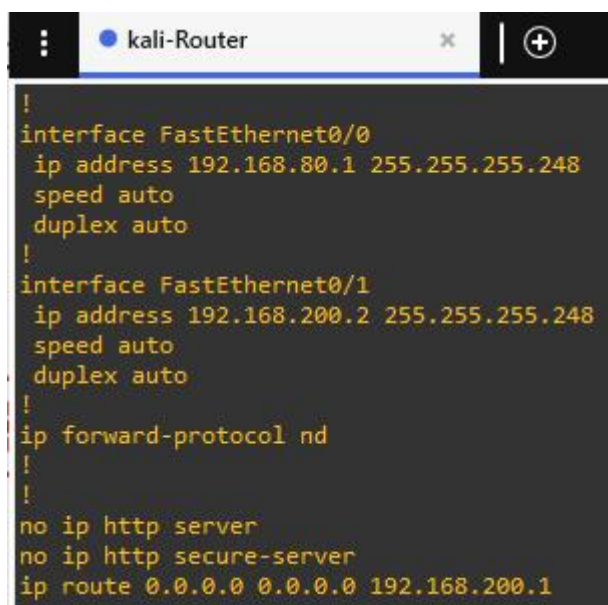
Description

Figure 9: Branch-route to kali network



```
ISP-Router
!
!
interface FastEthernet0/0
 ip address 192.168.163.1 255.255.255.0
 speed auto
 duplex auto
!
interface FastEthernet0/1
 ip address 192.168.71.1 255.255.255.0
 speed auto
 duplex auto
!
interface FastEthernet2/0
 ip address 192.168.200.1 255.255.255.248
 duplex full
!
interface GigabitEthernet3/0
 no ip address
 shutdown
 negotiation auto
!
ip forward-protocol nd
!
!
no ip http server
no ip http secure-server
ip route 192.168.10.0 255.255.255.0 192.168.163.10
ip route 192.168.20.0 255.255.255.0 192.168.71.10
ip route 192.168.80.0 255.255.255.248 192.168.200.2
!
```

Figure 10: ISP-Router configuration



```
!
interface FastEthernet0/0
 ip address 192.168.80.1 255.255.255.248
 speed auto
 duplex auto
!
interface FastEthernet0/1
 ip address 192.168.200.2 255.255.255.248
 speed auto
 duplex auto
!
ip forward-protocol nd
!
!
no ip http server
no ip http secure-server
ip route 0.0.0.0 0.0.0.0 192.168.200.1
!
```

Figure 11: kali-Router configuration

3.1.1.6 NAT Configuration

NAT was configured differently on both firewalls to support normal outbound traffic while keeping the IPsec tunnel free of translation. The Branch firewall used Hybrid NAT mode, which allowed OPNsense to apply automatic NAT for internet-bound traffic, while also permitting manual rules to override specific behavior when needed.

A manual No NAT rule was added for the network 192.168.20.0/24 (Branch LAN) when communicating with 192.168.10.0/24 (HQ LAN). This ensured that traffic passing through the IPsec tunnel kept its original internal IP addresses instead of being translated. Without this rule, the tunnel would fail to route packets correctly, and devices in both LANs would not be able to communicate.

On the HQ firewall, default automatic NAT was sufficient because the Branch firewall handled the necessary exemption. As a result, both LANs were able to exchange traffic normally through the encrypted VPN tunnel, while NAT continued to operate for regular outbound connections.



Figure 12: Branch NAT & Branch NAT rule

3.1.1.7 IDS/IPS Rule Configuration (Suricata)

Suricata IDS/IPS was configured on the HQ firewall to monitor both the LAN and WAN interfaces and provide real-time threat detection and prevention. IPS mode and promiscuous mode were enabled to allow deep inspection of traffic and automatic blocking of malicious packets. The Hyperscan pattern matcher was selected to improve performance and speed during signature processing.

A variety of rulesets were activated, including exploit, DoS, and malware categories from the Emerging Threats feed. These rules detect common attack patterns such as protocol misuse, DoS attempts, reconnaissance scans, and trojan-related activity. Many of the rules were set to alert, while others—especially high-risk signatures—were configured to drop malicious traffic automatically.

During testing, Suricata successfully generated alerts for attempted DoS and exploit-related signatures, and the HQ firewall showed these alerts in real time. This confirmed that the IDS/IPS was inspecting traffic correctly and responding to potential threats.

☐ sid	Action	Source	ClassType	Message	Info / Enabled
☐ 2000005	alert	emerging-exploit.rules	attempted-dos	ET EXPLOIT Cisco Telnet Buffer Overflow	☒
☐ 2000006	alert	emerging-dos.rules	attempted-dos	ET DOS Cisco Router HTTP DoS	☒
☐ 2000007	alert	emerging-exploit.rules	attempted-dos	ET EXPLOIT Catalyst SSH protocol mismatch	☒
☐ 2000009	alert	emerging-deleted.rules	attempted-dos	ET DELETED Cisco IOS HTTP DoS	☒
☐ 2000010	alert	emerging-dos.rules	attempted-dos	ET DOS Cisco 514 UDP flood DoS	☒
☐ 2000011	alert	emerging-dos.rules	attempted-dos	ET DOS Catalyst memory leak attack	☒
☐ 2000012	alert	emerging-deleted.rules	attempted-dos	ET DELETED Cisco %u IDS evasion	☒
☐ 2000013	alert	emerging-deleted.rules	attempted-dos	ET DELETED Cisco IOS HTTP server DoS	☒
☐ 2000015	alert	emerging-p2p.rules	trojan-activity	ET P2P Phatbot Control Connection	☒
☐ 2000016	alert	emerging-deleted.rules	attempted-dos	ET DELETED SSL Bomb DoS Attempt	☒

Showing 1 to 10 of 208948 entries ☐ Alert Drop

Figure 13: Suricata rules and alerts displayed on the HQ firewall

advanced mode

General Settings

Enabled ☒

IPS mode ☒

Promiscuous mode ☒

Interfaces

LAN, WAN

Clear All Select All

Detection

Pattern matcher

Hyperscan

Logging

Enable syslog alerts ☒

Enable eve syslog output ☐

Rotate log

Weekly

Save logs

4

Figure 14: IDS/IPS general settings showing IPS mode enabled on both LAN and WAN interfaces

3.1.1.8 Attack Surface Reduction (Hardening)

Some basic hardening steps were applied on the HQ firewall to reduce unnecessary access. The most important change was controlling what traffic is allowed through the IPsec tunnel. Only the two LAN networks were permitted to communicate with each other, and a final block rule was added to stop any other IPsec traffic. This keeps the tunnel limited to what the network actually needs and prevents unwanted connections.

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Other small hardening measures included using OPNsense's default protections on the WAN interface and restricting management access to the LAN only. These steps helped make the HQ firewall more secure without adding too much complexity.



☐		IPv4 ICMP	192.168.10.0/24	*	192.168.20.0/24	*	*	*	Allow ICMP from HQ LAN to Branch LAN			
☐		IPv4 ICMP	192.168.20.0/24	*	192.168.10.0/24	*	*	*	Allow ICMP from Branch LAN to HQ LAN			
☐		IPv4 TCP/UDP	192.168.10.0/24	*	192.168.20.0/24	*	*	*	Allow HQ LAN to Branch LAN traffic			
☐		IPv4 TCP/UDP	192.168.20.0/24	*	192.168.10.0/24	*	*	*	Allow Branch LAN to HQ LAN traffic			
☐		IPv4 *	*	*	*	*	*	*	Block all non-authorized IPsec traffic			

Figure 15: HQ Firewall rules

3.1.1.9 Logging and Monitoring Setup

The HQ firewall was configured to log and monitor both security events and network activity. Suricata-generated alerts were already being logged to identify which packets had been detected as suspicious or which were blocked when testing was performed, such as using nmap -A. In addition, the firewall rules were set to log any traffic based on configured policies, confirming whether that traffic was either allowed or denied.

System performance, currently active connections; and the state of the IPsec tunnel are fully monitored with the native OPNsense interface as well as its built-in reporting capabilities. The OPNsense interface has everything needed to view the current configuration of the IPsec tunnel, the status of the IDS/IPS as well as how the network performs in both normal and attack situations. Overall, the complete monitoring of the network can be maintained without adding any external monitoring products.

3.1.1.10 Final NGFW Policy Functionality

With all configuration completed, the HQ Firewall has now become a fully functioning integrated system made up of different security features brought together in one place. The Firewall Policies allow regulated traffic through, while the Intrusion Detection/Intrusion Prevention System (IDS/IPS) was used to monitor all incoming/outgoing traffic and stop anything that was identified as malicious or potentially harmful. The Virtual Private Network (VPN) creates an encrypted communication channel between the HQ and the Branch offices, while the Domain Name System (DNS) filtering limits the user's ability to access potentially dangerous or unwanted web sites addresses. All these components of the firewall work together seamlessly to provide the best possible protection to the entire HQ network while at the same time continuing to allow normal access by all users.

3.1.2 Additional Security Services

Besides basic firewalling and VPN connectivity, the HQ firewall also included several additional security services to enhance network protection. Suricata IDS/IPS was installed and configured to inspect both WAN and LAN traffic, using updated rule sets such as abuse.ch and Feodo Tracker to detect malware and command-and-control activity. DNS-based filtering was also enabled through Unbound DNS to block unwanted or suspicious domains.

3.1.2.1 Installed Plugins

Only a few essential plugins were installed on the HQ OPNsense firewall to support the security features required for the project. The most important component was the Suricata IDS/IPS engine, which is included by default in OPNsense but operates as a core service once enabled. This plugin allowed the firewall to apply updated threat intelligence rules, inspect live traffic, and block malicious activity.

3.1.2.2 Rulesets (abuse.ch, ET Open)

Several IDS/IPS rulesets were made available on the HQ firewall, including abuse.ch and ET Open. These rule feeds were later used in the IDS/IPS configuration to provide threat signatures for malware and command-and-control activity. The detailed activation and behavior of these rules are explained in section 5.3.1.7.

3.1.2.3 DNS Resolver & DNSBL

To prevent users from accessing unwanted websites before any type of connection is established, Unbound DNS is used in conjunction with a DNSBL (Domain Name Service Block-List) on the HQ Firewall. Facebook.com and Youtube.com were placed on the "blocklist". Conversely, other domains were placed on the "whitelist" (the permitted list). The blocked domains resolve to 0.0.0.0. Thus, access to those unwanted sites is denied by that address. This is an effective way to control content through DNS Filtering, which does not require installation of a complete Web Filtering

engine.

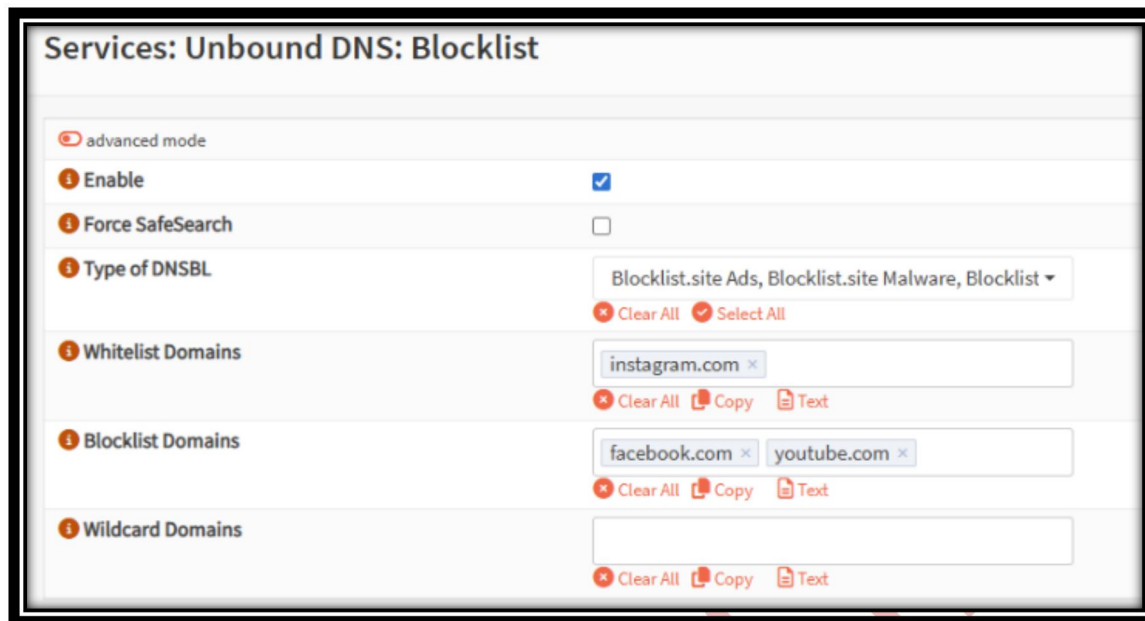


Figure 16: DNS Blocklist

3.1.2.4 Web Filtering Policies

The WEB Filtering function was implemented with DNSBL on the HQ firewall in Unbound DNS; however, instead of using a complete system only domain filtering will be done for restricting the use of certain sites. The level of complexity added in the NGFW was just enough to show that it can still be utilized effectively to control what users are spending time on online without having to make things overly difficult for users or create additional overhead.

3.1.2.5 Backup & Configuration Management

To ensure consistency during the implementation of this project, both the Headquarters (HQ) and Branch Office (BO) OPNsense Installations took regular backups of the firewall configuration as the setup continued. OPNsense has a backup function that allows the user to create a backup of the entire configuration file as an XML file. This makes it simple for users to restore their systems if they experience an error or a misconfiguration on their part.

Regular backups were created when milestone achievements occurred on the project.

3.1.3 Remote Connectivity System

3.1.3.1 IPsec Tunnel Establishment and Testing

The tunnels were established between the HQ and the Branch following successful configuration of Phase I and II on each firewall, with both tunnels showing an INSTALLED

status according to the Status page of each Firewall's Web Interface. There were no issues in authenticating, negotiating and exchanging keys from either end of the connection.

VPN: IPsec: Status Overview

Phase 1

Q

Search

7

☒	Status	Connection	Version	Local ID	Local IP	Remote ID	Remote IP	Time	Bytes in	Bytes out	
☒		HQ-to-Branch	IKEv2	192.168.163.10	192.168.163.10	192.168.71.10	192.168.71.10	1304			x Q

1

Showing 1 to 1 of 1 entries

Phase 2

Q

Search

7

☐	Local subnets	spi-in	spi-out	Remote subnets	State	Time	Bytes in	Bytes out
☐	192.168.10.0/24	c8a0533c	ca02617c	192.168.20.0/24	INSTALLED	1305		

Figure 17: HQ-IPsec Status

To verify that the tunnels were established successfully as well as routing traffic properly, connectivity tests were conducted via the LAN of both locations with the use of the ICMP protocol. An example of this would be that a Computer on the LAN of HQ addressed 192.168.10.0/24 successfully ICMP Pinged a machine on the LAN of the Branch Location addressed to 192.168.20.0/24. This was expected in that it would also show a successful ICMP Ping from the Branch Location back to its Headquarters providing further proof that the encrypted tunnels were properly forwarding traffic to/from each location.

```
PC1> ip dhcp
DORA IP 192.168.10.63/24 GW 192.168.10.1

PC1> ping 192.168.20.51

84 bytes from 192.168.20.51 icmp_seq=1 ttl=62 time=20.831 ms
84 bytes from 192.168.20.51 icmp_seq=2 ttl=62 time=19.071 ms
84 bytes from 192.168.20.51 icmp_seq=3 ttl=62 time=18.832 ms
84 bytes from 192.168.20.51 icmp_seq=4 ttl=62 time=19.477 ms
84 bytes from 192.168.20.51 icmp_seq=5 ttl=62 time=19.483 ms
```

Figure 18: ping from HQ-PC1(192.168.10.63) to Branch-PC4 (192.168.20.51)

```
PC4> ip dhcp
DDORA IP 192.168.20.51/24 GW 192.168.20.1

PC4> ping 192.168.10.63

84 bytes from 192.168.10.63 icmp_seq=1 ttl=62 time=18.819 ms
84 bytes from 192.168.10.63 icmp_seq=2 ttl=62 time=19.415 ms
84 bytes from 192.168.10.63 icmp_seq=3 ttl=62 time=19.377 ms
84 bytes from 192.168.10.63 icmp_seq=4 ttl=62 time=19.415 ms
84 bytes from 192.168.10.63 icmp_seq=5 ttl=62 time=19.866 ms
```

Figure 19: ping from Branch-PC4 to HQ-PC1

During testing the tunnels were stable as indicated by the incremental increase of the byte counters shown on each firewall's IPsec Status page each time there was traffic between the two Site Locations, indicating that the VPN would be completely functional as anticipated providing securely routed network connections for each of the two Site Locations.

3.1.4 Dashboard and Monitoring Configuration

The OPNsense dashboard was customized to show the most important information needed during the project. Widgets such as system health, interface statistics, firewall logs, and IPsec status were added to make it easier to monitor the network in real time. This allowed quick visibility into CPU and memory usage, active connections, and whether the VPN tunnel was up, in addition to the dashboard.

3.1.5 Web Management Interface

Both firewalls in the project were managed through OPNsense's web-based interface, which made configuration and monitoring easy throughout the setup. Access to the GUI was limited to the LAN interface for security reasons, ensuring that no external device could reach the management portal from the WAN

3.1.6 Network Implementation

3.1.6.1 Overview

The last network configuration mimics a typical enterprise environment using a multi-site structure. Each of the HQ and Branch Networks benefits from being connected through a VPN tunnel created by connecting both networks through an ISP router (i.e., the "Public" part of this Enterprise network).

The Kali Linux Network is structured on its own Network and is used as a Security Testing tool. The Kali machine is attached to its own dedicated Router, and has Internet access via the ISP Router, but is not connected directly to either of the Firewalls. By designing it this way, we can authentically simulate real-world attacks from sources that are unknown or untrusted.

Each site operates on its own LAN subnet:

- HQ LAN: 192.168.10.0/24
- Branch LAN: 192.168.20.0/24

Local PCs connect through switches at each site, while secure communication between HQ and Branch is handled through the IPsec VPN tunnel. All traffic between sites passes through the ISP router, which ensures realistic routing behavior and a clear separation between internal and external networks.

Overall, this implementation successfully integrates routing, VPN, NGFW inspection, and external attack simulation into one coherent test environment.

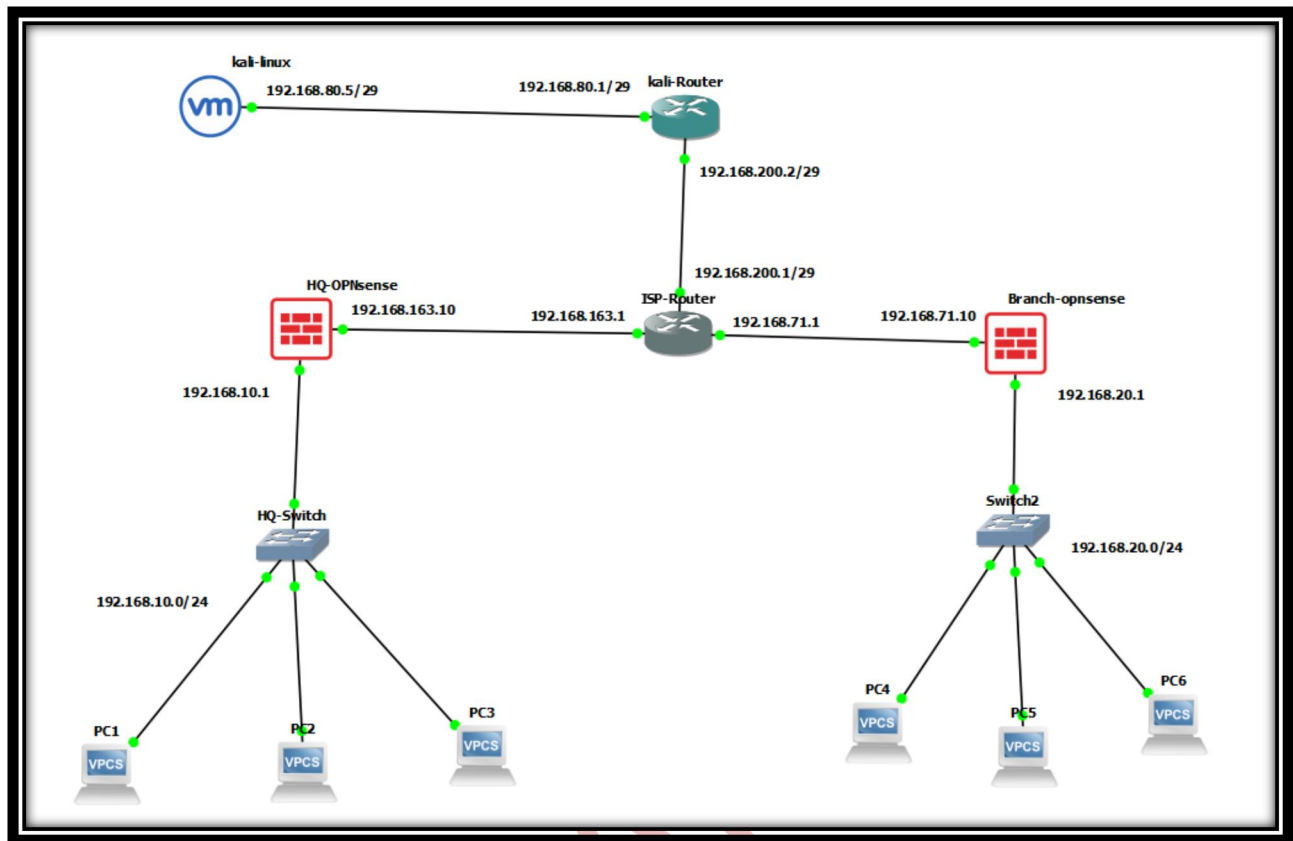


Figure 20: Network topology

3.1.6.2 Device Roles (HQ Firewall, Branch Firewall, ISP Router, Kali)

Four devices in the network deliver their function in regard to providing security, routing, and testing throughout the project:

1. The first is the HQ Firewall (OPNsense).

The HQ firewall is the main firewall in the topology. It provides LAN routing, DHCP services, NAT, and all core security functions such as IDS/IPS, DNS filtering, & Access Control Policies. The HQ firewall is also one end of the IPsec VPN tunnel connecting the HQ Network and the Branch Network, providing a secure connection.

2. The second is the Branch Firewall (OPNsense).

The Branch firewall is a mirror image of the HQ firewall, with less security than the HQ firewall. The Branch firewall disables IDS/IPS on purpose to compare protected and unprotected sites' responses when attacked. The Branch Firewall is responsible for providing LAN services on an

established Branch network, DHCP services to the Branch Network, and the other endpoint of the IPsec VPN Tunnel connecting the HQ & Branch Networks.

3. The ISP Router

is primarily responsible for connecting both OPNsense Firewalls to the outside world. The ISP router emulates the Public Internet and routes all data and connection requests from the HQ & Branch Networks back through the ISP router to the Kali Network. Communication (including IPsec negotiations) always passes through the ISP Router, which separates the internal from the external networks in a realistic way.

4. Kali Linux & Kali Router

The Kali machine is used for penetration testing, scanning, and traffic generation. It does not connect directly to the HQ or Branch firewalls. Instead, it connects through its own router, which then links to the ISP router. This setup allows Kali to behave like an external attacker or untrusted client, providing accurate testing conditions for IDS/IPS evaluation.

3.1.6.3 HQ Network Setup and Interfaces

The HQ firewall was configured with two main interfaces: a WAN interface connected to the ISP router (192.168.163.10) and a LAN interface serving the internal network (192.168.10.1/24). These interfaces were used throughout the project for routing, DHCP services, and enforcing NGFW security policies.

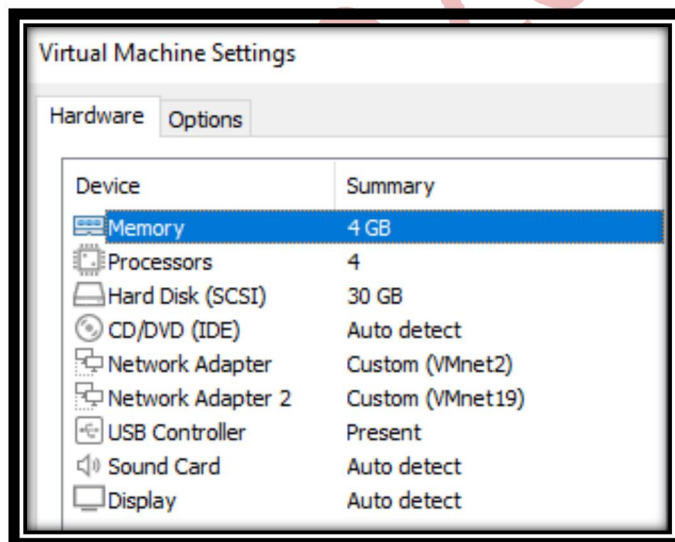


Figure 21: HQ firewall setup

3.1.6.4 Branch Network Setup and Interfaces

The Branch firewall was configured with two interfaces: a WAN interface connected to the ISP router (192.168.71.10) and a LAN interface for the Branch network (192.168.20.1/24). This setup provided basic routing, DHCP services, and the endpoint for the IPsec tunnel linking the Branch to the HQ network.

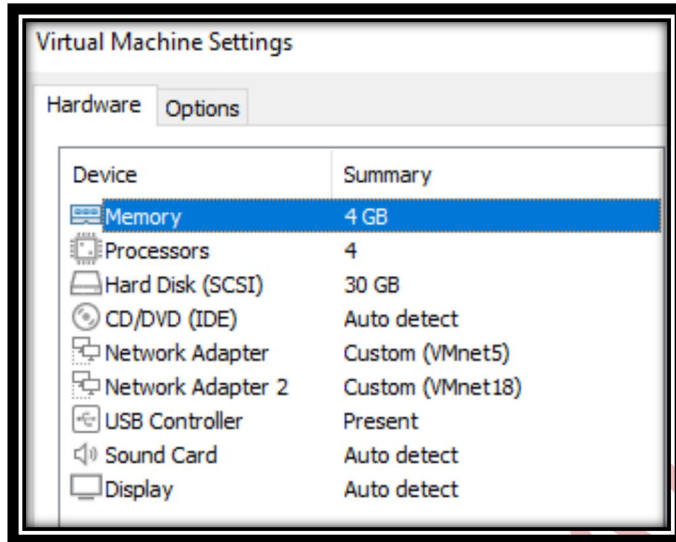


Figure 22: Branch firewall setup

3.1.6.5 Connectivity Notes

To achieve sound communication and continuity between the various topology segments, it was necessary to link all virtual machines and firewalls properly with the corresponding virtual network adapters during the installation of GNS3.

To facilitate this, two VMware network interfaces were assigned to each of the two OPNsense firewalls. The firewalls were assigned both a WAN and a LAN interface. GNS3 mapped these interfaces to their respective VMware VMnet networks.

For example, the HQ firewall was mapped to the WAN side through VMnet19 and LAN with VMnet2, while the Branch firewall was mapped to the WAN side through VMnet18 and LAN with VMnet5.

By ensuring that each VMware and GNS3 interface naming convention was the same throughout, testing of the IPsec tunnel and external scanning from Kali would be made extremely simple. When all network interface settings were correctly configured, the GNS3 network remained stable, and end-to-end connectivity was achieved.

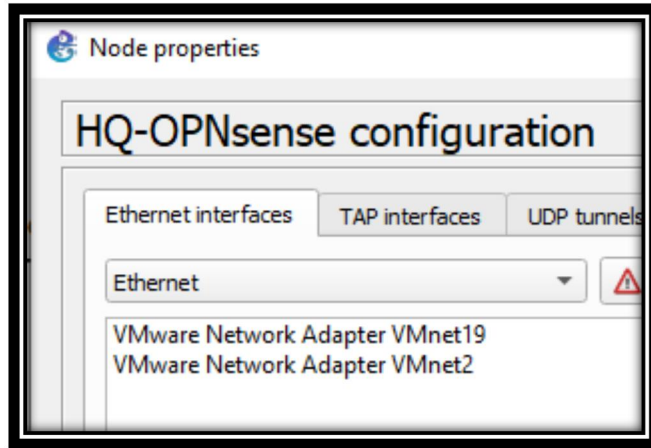


Figure 23: HQ-OPNsense VMware Network Adapter Mapping in GNS3

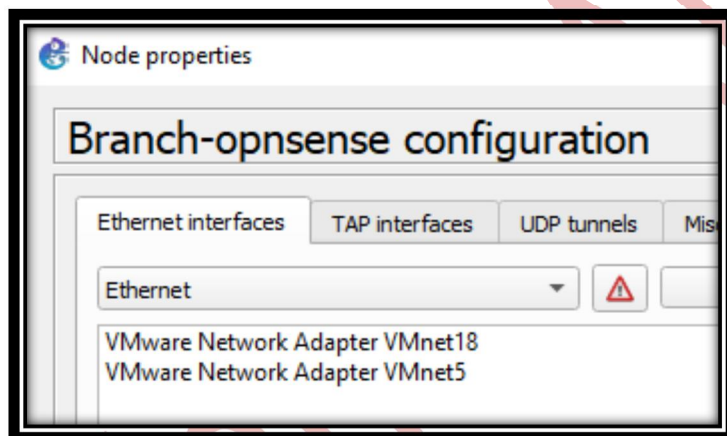


Figure 24: Branch-OPNsense VMware Network Adapter Mapping in GNS3

3.1.6.6 Challenges and Solutions

1. Interface Mapping Between VMware and GNS3

One of the first issues was matching each VMnet adapter to the correct interface inside GNS3. Incorrect mapping caused the firewalls to receive traffic on the wrong interface, leading to routing failures.

Solution: Each OPNsense VM was checked manually in VMware, and the adapters were matched to the correct VMnet numbers in GNS3. Once aligned, routing and DHCP began working normally.

2. IPsec Tunnel Failing to Come Up

Initially, the IPsec tunnel did not establish because one side used a wrong subnet mask and mismatched Phase 2 networks.

Solution: Both Phase 1 and Phase 2 settings were rechecked. After correcting the subnet definitions and PSK, the tunnel established successfully and showed INSTALLED on both firewalls.

3. No IDS/IPS Alerts During Testing

Suricata was enabled but did not trigger alerts during early tests. This was caused by placing the Kali machine in a different network path that the IDS was not monitoring.

Solution: IDS/IPS was enabled on both LAN and WAN interfaces of the HQ firewall, which allowed nmap scans and other tests to be detected properly.

4. Connectivity Drops Between HQ and Branch

Ping tests between LAN subnets sometimes failed because NAT rules interfered with IPsec traffic.

Solution: A “No NAT” rule was added for traffic between 192.168.10.0/24 and 192.168.20.0/24, ensuring clean routing through the tunnel.

3.1.6.7 Final Network Topology

The final topology represents the completed version of the network after configuring all routing, security policies, IDS/IPS, DNS filtering, and the IPsec VPN. All devices are connected according to the design, and end-to-end communication works correctly between the HQ and Branch LANs. The diagram below summarizes how the environment looks after full implementation.

Device	Type	Interface	IP Address	Subnet	Description
Kali Linux	Host	eth0	192.168.80.5	/29	Attack/testing machine
Kali Router	Router	fa0/0	192.168.80.1	/29	Connects Kali to ISP router
Kali Router	Router	fa0/1	192.168.200.2	/29	Link toward ISP router
ISP Router	Router	fa0/0	192.168.200.1	/29	Connects to Kali Router
ISP Router	Router	fa0/1	192.168.163.1	/24	WAN gateway for HQ firewall
ISP Router	Router	fa0/2	192.168.71.1	/24	WAN gateway for Branch firewall
HQ OPNsense Firewall	NGFW	WAN (em1)	192.168.163.10	/24	WAN side toward ISP router
HQ OPNsense Firewall	NGFW	LAN (em0)	192.168.10.1	/24	HQ internal network gateway
Branch OPNsense Firewall	NGFW	WAN (em1)	192.168.71.10	/24	WAN side toward ISP router
Branch OPNsense Firewall	NGFW	LAN (em0)	192.168.20.1	/24	Branch LAN gateway

Figure 25: Full Network IP Addressing Summary

Network	Subnet	Devices Included	Purpose
Kali Network	192.168.80.0/29	Kali Linux + Kali Router	External attacker simulation
Kali-ISP WAN Link	192.168.200.0/29	Kali Router ↔ ISP Router	Internet-simulation path
HQ WAN	192.168.163.0/24	HQ Firewall + ISP Router	Public-facing WAN side
Branch WAN	192.168.71.0/24	Branch Firewall + ISP Router	Public-facing WAN side
HQ LAN	192.168.10.0/24	HQ PCs + HQ Firewall	Internal secure LAN
Branch LAN	192.168.20.0/24	Branch PCs + Branch Firewall	Internal secure LAN

Figure 26: Network Segments Summary

3.2 Testing and Validation

After completing the full configuration of the NGFW environment, several tests were performed to validate connectivity, security features, and overall network behavior. The goal of this stage was to confirm that all components—routing, VPN, IDS/IPS, DNS filtering, and firewall policies—were working as intended and providing the expected level of protection.

3.2.1 IDS/IPS Detection Test (Nmap Scan)

To validate the effectiveness of the IDS/IPS system, an nmap -A scan was launched from the Kali machine targeting the HQ firewall's WAN interface (192.168.163.10). The scan attempted OS detection, service enumeration, and aggressive probing, which are typical behaviors of reconnaissance attacks.

```
(kali㉿kali)-[~]
$ nmap -A 192.168.71.10
Starting Nmap 7.95 ( https://nmap.org ) at 2025-12-04 02:54 EST
Stats: 0:00:27 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 33.33% done; ETC: 02:56 (0:00:44 remaining)
Nmap scan report for 192.168.71.10
Host is up (0.038s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE  VERSION
53/tcp    open  domain   (generic dns response: REFUSED)
80/tcp    open  http     OPNsense
| fingerprint-strings:
|   FourOhFourRequest:
|     HTTP/1.0 301 Moved Permanently
|     Location: https:///nice%20ports%2C/Trinity.txt.bak
|     Content-Length: 0
|     Connection: close
|     Date: Thu, 04 Dec 2025 10:55:08 GMT
|     Server: OPNsense
|   GenericLines:
|     HTTP/1.0 400 Bad Request
|     Content-Type: text/html
|     Content-Length: 345
|     Connection: close
|     Date: Thu, 04 Dec 2025 10:55:08 GMT
|     Server: OPNsense
|     <?xml version="1.0" encoding="iso-8859-1"?>
|     <!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN"
|     "http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">
|     <html xmlns="http://www.w3.org/1999/xhtml" xml:lang="en" lang="en">
|     <head>
|     <title>400 Bad Request</title>
|     </head>
```

Figure 27: nmap attack

The IDS/IPS on the HQ firewall successfully detected the scan. Suricata generated multiple alerts showing clear evidence of the attack, including entries classified as “ET SCAN Possible Nmap User-Agent Observed”. The logs displayed repeated blocked events originating from the Kali IP (192.168.80.5), confirming that the firewall not only recognized the attack but applied IPS actions to drop the traffic.

Services: Intrusion Detection: Administration

We strongly advise to use policies instead of single rule based changes to limit the size of the configuration. A list of all manual changes can be revised in the policy editor (available here)

Settings Download Rules User defined Alerts Schedule

Search 2025/12/04 2:48 50

Timestamp	SID	Action	Interface	Source	Port	Destination	Port	Alert	Info
2025-12-04T10:26:01.304718+0300	2024364	blocked	em1	192.168.80.5	44318	192.168.163.10	80	ET SCAN Possible Nmap User-Agent Observed	
2025-12-04T10:26:01.304718+0300	2024364	blocked	em1	192.168.80.5	44318	192.168.163.10	80	ET SCAN Possible Nmap User-Agent Observed	
2025-12-04T10:26:01.107031+0300	2024364	blocked	em1	192.168.80.5	44308	192.168.163.10	80	ET SCAN Possible Nmap User-Agent Observed	
2025-12-04T10:26:01.107031+0300	2024364	blocked	em1	192.168.80.5	44308	192.168.163.10	80	ET SCAN Possible Nmap User-Agent Observed	
2025-12-04T10:26:00.846716+0300	2024364	blocked	em1	192.168.80.5	44256	192.168.163.10	80	ET SCAN Possible Nmap User-Agent Observed	
2025-12-04T10:26:00.846716+0300	2024364	blocked	em1	192.168.80.5	44256	192.168.163.10	80	ET SCAN Possible Nmap User-Agent Observed	
2025-12-04T10:26:00.846716+0300	2024364	blocked	em1	192.168.80.5	44230	192.168.163.10	80	ET SCAN Possible Nmap User-Agent Observed	
2025-12-04T10:26:00.846716+0300	2024364	blocked	em1	192.168.80.5	44230	192.168.163.10	80	ET SCAN Possible Nmap User-Agent Observed	
2025-12-04T10:26:00.701276+0300	2024364	blocked	em1	192.168.80.5	44208	192.168.163.10	80	ET SCAN Possible Nmap User-Agent Observed	
2025-12-04T10:26:00.701276+0300	2024364	blocked	em1	192.168.80.5	44208	192.168.163.10	80	ET SCAN Possible Nmap User-Agent Observed	
2025-12-04T10:26:00.648786+0300	2024364	blocked	em1	192.168.80.5	44188	192.168.163.10	80	ET SCAN Possible Nmap User-Agent Observed	
2025-12-04T10:26:00.648786+0300	2024364	blocked	em1	192.168.80.5	44188	192.168.163.10	80	ET SCAN Possible Nmap User-Agent Observed	
2025-12-04T10:25:54.144752+0300	2024364	blocked	em1	192.168.80.5	44170	192.168.163.10	80	ET SCAN Possible Nmap User-Agent Observed	
2025-12-04T10:25:54.144752+0300	2024364	blocked	em1	192.168.80.5	44170	192.168.163.10	80	ET SCAN Possible Nmap User-Agent Observed	
2025-12-04T10:25:54.092628+0300	2024364	blocked	em1	192.168.80.5	44160	192.168.163.10	80	ET SCAN Possible Nmap User-Agent Observed	
2025-12-04T10:25:54.092628+0300	2024364	blocked	em1	192.168.80.5	44160	192.168.163.10	80	ET SCAN Possible Nmap User-Agent Observed	

Showing 116 of 34 entries

Figure 28: Suricata IDS/IPS alerts triggered by the Nmap aggressive scan from Kali

These results demonstrate that the NGFW correctly identifies external scanning attempts and responds by blocking the malicious packets. As expected, the Branch firewall—where IDS/IPS was intentionally disabled—did not generate any alerts during the same scan, highlighting the difference in protection levels between the two sites

Branch Firewall

Without IPS/IDS, Nmap was able to gather significantly more data, including:

- Open ports: 53, 80, 443
- HTTP and HTTPS server details
- Full HTML responses
- Cookies and CSRF tokens
- Redirect behaviour
- SSL certificate details
- Accurate OS fingerprint
- Full-service banners

Result:

Extensive information was disclosed. Attack succeeded

3.2.2 Management Interface Test

The management interface was tested by accessing the OPNsense web GUI from inside each LAN. Both firewalls were reachable only through their LAN interfaces, confirming that remote access from the WAN was properly blocked. All pages loaded normally, and configuration changes were applied without issues, verifying that the management interface was secure and functioning correctly.

3.2.3 Integration Test

All network components were tested together, and the system worked as expected. HQ and Branch LANs communicated through the VPN, DNS filtering applied correctly, and IDS/IPS detected scans without affecting normal traffic.

3.2.4 Performance & Reliability

During testing, the firewalls remained stable, and all services—including VPN, DHCP, and IDS/IPS—performed without interruptions. CPU and memory usage stayed within normal limits, and no crashes or connectivity issues were observed, confirming the system's reliability.

3.3 Summary

Chapter 4

This chapter presented the full design and implementation of the NGFW setup, including routing, VPN configuration, IDS/IPS, DNS filtering, and network deployment. All components worked together as expected, and testing confirmed secure communication, successful threat detection, and stable system performance.

4. Chapter 5: Results and Discussion

4.1 Findings

Results indicated a superior amount of visibility available by an NGFW to help detect more types of threats than were detected by traditional firewalls. The combination of IDS/IPS and Site-to-Site VPN security effectively blocked most scanning activity while still ensuring secure communication between the two sites. A DNS filter within an NGFW solution, allowed for an easy method for adding content filtering with little configuration.

4.2 Goals Achieved

The project successfully achieved its main goals: establishing secure communication through IPsec, detecting external threats using IDS/IPS, and applying basic content filtering with DNSBL. The NGFW setup demonstrated effective protection, stable performance, and clear improvements over a basic firewall configuration.

4.3 Further Work

Future work could include enabling Deep Packet Inspection (DPI) for better traffic visibility, adding full web filtering, or expanding the setup to include more networks to test scalability.

4.4 Importance of System Testing

System testing is essential to confirm that security features work as expected, detect weaknesses early, and ensure the NGFW performs reliably under real network conditions.

4.5 Ethical, Legal, and Social Considerations

Utilizing security applications demands a high degree of responsibility and ethics to ensure that testing takes place only on sanctioned systems. Adherence to legal guidelines is also necessary as performing unauthorized scans or capturing data could produce violations of numerous regulations. In addition, utilizing sufficient security safeguards would help preserve the privacy of users and lessen the potential for improper or dangerous acts against consumers/victims.

5. Chapter 6: Conclusions

The project demonstrated how an NGFW can connect networks securely, detect threats, and control traffic through its use of features such as IPsec, IDS/IPS, and DNS filtering. Testing results show that the NGFW solution had reliable performance for the system and continued to provide significant benefits compared to basic firewall solutions. In addition, the implementation of the NGFW solution provides confirmation of the effectiveness of using layered security approaches to secure modern networks.

6. References

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